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$$\begin{aligned}& \int (x^3 - x^2) \cos(x) dx \\& \left\{ \begin{array}{l} u = x^3 - x^2 \\ dv = \cos(x) dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 3x^2 - 2x dx \\ v = \sin(x) \end{array} \right. \\& = (x^3 - x^2) \sin(x) - \int (3x^2 - 2x) \sin(x) dx \\& \left\{ \begin{array}{l} u = 3x^2 - 2x \\ dv = \sin(x) dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = (6x - 2) dx \\ v = -\cos(x) \end{array} \right. \\& = (x^3 - x^2) \sin(x) + (3x^2 - 2x) \cos(x) - \int (6x - 2) \cos(x) dx \\& \left\{ \begin{array}{l} u = 6x - 2 \\ dv = \cos(x) dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 6 dx \\ v = \sin(x) \end{array} \right. \\& = (x^3 - x^2) \sin(x) + (3x^2 - 2x) \cos(x) - (6x - 2) \sin(x) - 6 \cos(x) + C\end{aligned}$$

References